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Robust Neuro-sliding Mode Control for Position Servo-actuated Robot Manipulators

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Abstract: Artificial neural networks (ANN) are well known for their powerful function approximation capabilities, which have been extensively utilized in control systems to model complex dynamic behaviors. Specifically, in robotics, neurocontrol schemes have been developed to address model nonlinearities. However, these models typically assume that robots receive torque as control inputs to the actuators. This assumption does not always hold true, as in applications requiring high-precision movements, most robotic manipulators are equipped with position servo actuators. These actuators do not receive torque inputs; instead, they receive control signals directed to the actuators. This discrepancy creates a gap between the analysis and the implementation of control schemes for position-commanded robots. To address this gap, in this study, we propose a neurocontroller designed for trajectory tracking based in a position servo-actuated robot manipulator model. Lyapunov analysis is employed to guarantee the stability of the closed-loop trajectories, while the convergence of tracking errors is ensured by the underlying integral sliding mode. Finally, simulations validate the effectiveness of our proposed method.

Keywords: Neurocontrol, position actuators, robot manipulator, sliding mode, stability.

1. INTRODUCTION

It is well known that the dynamic equations of robot manipulators are highly coupled and nonlinear, making them difficult to obtain in practice. To address this challenge, the use of various control methods have been proposed to circumvent the computation of the exact models. One example is the well-stablished robust inverse dynamics control, which guarantees that all system trajectories remain uniformly ultimately bounded, despite the presence of errors in modeling [1,2]. However, this method still requires an approximation of the system model. Another approach is the adaptive control, where is sufficient to know the structure of the dynamics (rather than the exact parameter values), specifically the so-called regressor and parameters vector. The latter becomes an adaptive vector [3,4]. For high-dimensional robots (e.g., robots with more than three degrees of freedom), the computation of the regressor and the parameters vector is complex. To overcome the need for an explicit model or its structure (as in adaptive control), the approximation properties of neural networks (NN) have been exploited [5-8], and applied to control of robot manipulators [9-12]. Seminal works includes [13,14], where the application of NN for constructing model-free learning controllers for nonlinear systems is demonstrated, covering areas such as feedback linearization, feedforward control compensation, optimal control, and reinforcement learning.

Recent work in neurocontrol for robot manipulators includes [15], where an adaptive neurocontrol method is proposed, considering output constraints and uncertainties in the model. This approach employs a Barrier Lyapunov Function, ensuring that tracking errors converge to a compact set. In [16], two types of adaptive bias NN are presented to improve the approximation accuracy of the neural network for the dynamics, utilizing a simple controller structure and concluding uniform ultimate boundedness. Authors in [17] introduce a NN and a new performance index to optimize the gains from a PID controller. Positive results are demonstrated through simulations for various scenarios. An adaptive control based on radial basis NN is used in [18], where the NN approximates the dynamics and compensates it via the control input, combined with a quasiliding mode. A neural controller considering input saturation and joints constraints is proposed in [19], with the neural network approximating the nonlinear dynamics and ensuring UUB stability. Adaptive NN are employed in [20] to control closed architecture industrial manipulators, where the neural network approximates the unknown part of the inner controller. Finally, in [21], an adaptive NN

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is used to approximate the dynamic uncertainties based in composite learning, concluding semiglobally practical fixed-time stability of the closed-loop signals.

On the other hand, the combination of neural networks with sliding mode control has been proposed to add robustess. For instance, in [22], an adaptive sliding mode is proposed, with the model compensated by approximating a Fuzzy Wavelet NN for an industrial robot, resulting in system stability. Similarly, in [23], a Radial Basis Function NN is used to compensate the nonlinear dynamic, and a disturbance observer is incorporated into the control with an additional term to enforce sliding mode behavior, achieving tracking despite the discontinuous control signal. Another approach is presented in [24], where a prescribed performance control is introduced, using a NN to compensate for nonlinear dynamics, thereby ensuring the boundedness of tracking errors. In [25], a neurocontroller based on state feedback is suggested to approximate compensate the model, also an integral sliding mode is introduced to reduce the tracking errors. In [26], a NN is used to compensate the nonlinear systems without pretraining, and a saturation term is added to the control structure to induce a sliding mode, ensuring the convergence of tracking errors.

The works in [15-26] guarantee that the error trajectories either converge to a bounded set or achieve convergence by implementing first order sliding modes. However this approach generates non-smooth control signals causing the chattering phenomena. To overcome this issue, in [27] the combination of integral sliding mode (ISM) control and adaptive linear elements is proposed to guarantee the convergence of the tracking errors. A limitation of this method is that it only approximates and compensates for the linear portion of the robot's dynamics.

The majority of the previously mentioned research works, including those in [15-26], assume that the robot manipulators under study receive torque input commands. Specifically, the operation of the actuators they are equipped with is based on the principle of receiving torque signals at the input to produce joint movement at the output. The traditional mathematical model of n degrees-of-freedom (DoF) robot manipulators is used to describe the dynamics of the robotic systems, serving as the foundation for designing control systems, depending on the specific problem to be addressed. This broadly used and accepted model assumes that the torque requested by the actuators is delivered instantaneously, representing an idealized behavior that has not been verified in practice.

However, a class of robot manipulators commonly used in research and education are equipped with position actuators, which provide high positioning accuracy and respond to position inputs. Despite this, they are modeled as if the actuators were of the torque type, leading to a significant discrepancy between the characteristics and practical behavior of these robotic systems and their theoretical description. Recently, in [28], the first proposal for a dynamic model of robot manipulators equipped with position servo actuators was presented, along with a method to convert any torque-requesting control system to one that can be directly applied to robots with position servo actuators. However, the design of control systems for these type of robots represents an area with great potential for exploration.

Therefore, in this work, we present an improvement to the model introduced in [28], adding important factors that can affect the system's behavior, such as viscous friction and external perturbations. We also propose the first neuro-integral sliding mode (Neuro-ISM) tracking control for robot manipulators with position servo actuators. In contrast to [15-26], we guarantee the convergence of the tracking errors to the origin with a smooth control signal, and unlike [27], we use a Multi Layer Perceptron NN, which allows to approximate nonlinear dynamics, leading to a better approximation and compensation.

The rest of this paper is organized as follows: The dynamic model of a robot and servo-robot manipulators, properties of neural networks, as well as the problem statement, are presented in Section 2. Section 3 presents our model-free neurocontroller, including the stability analysis. Simulations and discussion of the results are presented in Section 4. Finally, concluding comments are presented in Section 5. For simplicity, the term "servo-robot" will be used in this document to refer to manipulator robots equipped with position servo actuators.

2. PRELIMINARIES AND PROBLEM STATEMENT

2.1. Torque input n-DoF robot manipulators dynamics

Robot manipulators with n-DoF, inelastic links and viscous friction at its joints, which receive torque commands at the input are usually modeled using the following highly nonlinear equation [4]:

$$H(q)\ddot{q}+C(q,\dot{q})\dot{q}+B\dot{q}+g(q)=\tau, \tag{1}$$

where $q \in \mathbb{R}^n$ is the vector of joint angles, $\dot{q} \in \mathbb{R}^n$ is the vector of joint velocities, $\ddot{q} \in \mathbb{R}^n$ is the vector of joint accelerations, and $\tau \in \mathbb{R}^n$ is the input vector of induced or applied torques. The remaining components of model (1) are listed below:

1) $H(q) \in \mathbb{R}^{n\times n}$ is the inertia matrix, which is symmetric and positive definite.
2) $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ is the matrix of centrifugal and Coriolis forces matrix, assumed to be computed using the Christoffel symbols of the first kind.
3) $B \in \mathbb{R}^{n\times n}$ is a diagonal, positive definite matrix containing the viscous friction coefficient for each joint.
4) $g(q) \in \mathbb{R}^n$ is the vector of forces due to gravity.

The mechanical structure of position servo-actuated robot manipulators, similar to that of torque-driven robots, consists of a sequence of links connected by rotational joints. These joints enable relative motion between the links, forming an open kinematic chain. Consequently, the model for robot manipulators with position servo actuators is derived from the traditional dynamic model (1).

2.2. Position servo actuators dynamics

Recognized for their ability to achieve precise positioning with high accuracy, servo actuators are increasingly applied in robotics due to their reduced size and affordable cost [29]. These compact servomotors convert input electrical energy u into controlled rotational motion θ, and are equipped with an electrically-driven permanent magnet direct current (PMDC) motor, a position sensor (typically a potentiometer) to measure the angular displacement of the servo's axis, a reduction transmission system, and an embedded electronic control board, see Fig. 1 for reference. Servomotors can be either digital or analog; however, this work focuses on analog position servos [30].

An analog position servo actuator is a closed-loop system in which the controller sends a signal to the servo motor, specifying the desired position. The motor's circuitry then powers the motor to move towards this position. The internal control system of the servo actuator, responsible for correcting positioning errors, typically employs a Proportional-dominant PID controller. In this configuration, the contributions of the integral and derivative gains are minimal and can, therefore, be excluded from the analysis. A set of n position servo actuators can be modeled as follows [28]:

$$\begin{aligned} &J_i\ddot{\theta}_i + \frac{(k_i k_{bi} + R_i b_{0i})}{R_i}\dot{\theta}_i + \eta_{mi}\cdot\frac{k_i k_{Pi}}{R_i}\theta_i \\ &= \eta_{mi}\cdot\frac{k_i k_{Pi}}{R_i}u_i - \eta_{mi}\cdot\tau_{di} - \tau_{\Gamma_i}. \end{aligned} \tag{2}$$

In model (2), for each servo actuator indexed by $i = 1, \cdots, n$, the input position command is denoted by $u_i \in \mathbb{R}$, while the output corresponds to $\theta_i \in \mathbb{R}$ which represents the angular position of the shaft. $\dot{\theta}_i \in \mathbb{R}$ and $\ddot{\theta}_i \in \mathbb{R}$ are the angular velocity and angular acceleration, respectively. The model consists of several constant parameters, including J_i for rotor inertia, R_i as the armature resistance, k_i is the torque constant, k_{bi} stands for the back electromotive force constant, k_{Pi} denotes the inner proportional control gain, and b_{0i} is the viscous friction coefficient. Additionally, τ_{di} are the disturbance torques arising from the mechanical structure, such as the link attached to the servo, $\tau_{\Gamma i}$ represents other bounded external disturbances, and η_m is the servo internal gear ratio, computed as

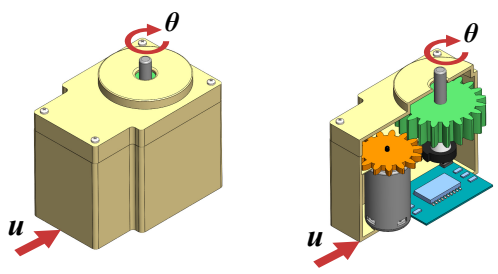


Fig. 1. Typical external and internal structure of a position servo actuator.

$$\eta_m = r_1/r_2, \tag{3}$$

where r_1 represents the number of teeth on the gear located at the rotor axis side, and r_2 denotes the number of teeth on the gear located at the output of the servo shaft. Note that, since the gear ratio in a position servo actuator is a reduction, the RPM of the PMDC motor decreases while increasing the output torque. Therefore, η_m is much less than 1 ($\eta_m \ll 1$).

The servo actuator model presented in (2) assumes that the armature inductance in the internal PMDC motor is negligible. This assumption is justified by the typically low inductance values found in PMDC motors. Moreover, the model presumes a direct mechanical connection between the structure of the robot manipulator and the position servo actuators, eliminating the need for gearboxes, couplings, belts, or pulley systems. As the servomotors directly drive the robot structure, the rotation of the shafts causes an immediate movement in the joints, meaning that $\theta_i = q_i$.

To facilitate subsequent analysis, and based on the fact that $\theta_i = q_i$, model (2) can be rewritten in the following compact vector form [28]:

$$H_{PS}\ddot{q} + B_{PS}\dot{q} + K_{PS}q = K_{PS}u - \Upsilon\tau_d - \tau_\Gamma, \tag{4}$$

where $q \in \mathbb{R}^n$ represents the output vector of joint positions, $u \in \mathbb{R}^n$ denotes the input, which is tipically a vector of desired positions, $H_{PS} = diag\ (J_1\ J_2\ \ldots\ J_n) \in \mathbb{R}^{n\times n}$, $B_{PS} = diag(\gamma_1\ \gamma_2\ \ldots\ \gamma_n) \in \mathbb{R}^{n\times n}$ and $K_{PS} \in \mathbb{R}^{n\times n} = diag(\sigma_1\ \sigma_2\ \ldots\ \sigma_n)$, where

$$\gamma_i \triangleq \frac{k_i k_{bi} + R_i b_{0i}}{R_i}, \quad \sigma_i \triangleq \eta_{mi}\cdot\frac{k_i k_{Pi}}{R_i}. \tag{5}$$

Furthermore, the total unknown disturbance $\tau_\Gamma \in \mathbb{R}^n$ is assumed to be continuously differentiable and bounded, i.e., $\|\tau_\Gamma\| \leq \epsilon_\Gamma$, $\tau_d \in \mathbb{R}^n$ represents the torque disturbances due to mechanical load, and $\Upsilon = diag(\eta_{m_1}\ \eta_{m_2}\ \ldots\ \eta_{m_n})$.

2.3. Servo-robot dynamics

The second-order nonlinear ordinary differential equation that represents the dynamic model of n-DoF position servo-actuated robot manipulators with rotational joint is given by the following expression:

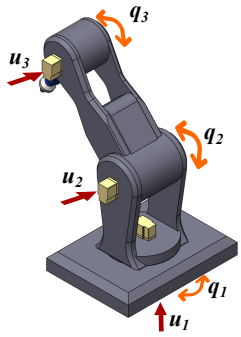


Fig. 2. Illustrative representation of a 3-DoF servo-robot manipulator.

$$H_{SR}(q)\ddot{q} + C_{SR}(q,\dot{q})\dot{q} + \Upsilon B\dot{q} + \Upsilon g(q) + K_{PS}q = K_{PS}u - \tau_\Gamma. \tag{6}$$

In (6), $q = (q_1, q_2, ..., q_n)^T \in \mathbb{R}^n$ denotes the vector of joint positions, while $u = (u_1, u_2, ..., u_n)^T \in \mathbb{R}^n$ represents the vector of position signals demanded to the actuators, $H_{SR} = \Upsilon \cdot H(q) + H_{PS}$, and $C_{SR}(q,\dot{q}) = \Upsilon \cdot C(q,\dot{q}) + B_{PS}$, where Υ is a diagonal matrix that depends on the internal gear ratio of the servos, $B \in \mathbb{R}^{n\times n}$ is the damping matrix, accounting for viscosity, and $g(q) \in \mathbb{R}^n$ stands for the vector of gravitational forces, $K_{PS} \in \mathbb{R}^{n\times n} = diag(\sigma_1\ \sigma_2\ ...\ \sigma_n)$, and $\tau_\Gamma \in \mathbb{R}^n$ is a vector of bounded external disturbances, which may arise from environmental factors such as electrical noise, vibrations, wind, or air currents, among other influences. Fig. 2 presents a 3-DoF robot manipulators actuated with position servomotors.

The dynamic model of servo-robots (6) is derived under the assumption that the mechanical structure of the robots is described by (1), and that this structure introduces a torque disturbance (τ_d) to the actuators due to mechanical load. Subsequently, τ_d is solved in (2) and substituted into (1), considering that the robot's inputs are the desired positions provided to the actuators.

2.4. Servo-robot model properties

The servo-robot dynamics, left-hand side of (6), can be linearly parametrizable as

$$H_{SR}(q)\ddot{q}_r + C_{SR}(q,\dot{q})\dot{q}_r + \Upsilon B\dot{q} + \Upsilon g(q) + K_{PS}q = Y_r\Theta, \tag{7}$$

where $\dot{q}_r$ is a nominal reference to be defined. Furthermore, multiplying both sides of (6) by K_{PS}^{-1} we obtain

$$K_{PS}^{-1}(H_{SR}(q)\ddot{q} + C_{SR}(q,\dot{q})\dot{q} + \Upsilon B\dot{q} + \Upsilon g(q) + K_{PS}q) = u - \tau_\Gamma. \tag{8}$$

Since K_{PS} is a diagonal matrix with positive constant parameters, the left-hand side of (8) is also linearly parametrizable in a nominal reference q_r as

$$K_{PS}^{-1}(H_{SR}(q)\ddot{q}_r + C_{SR}(q,\dot{q})\dot{q}_r + \Upsilon B\dot{q}_r + \Upsilon g(q) + K_{PS}q) = Y_{r2}\Theta_2. \tag{9}$$

Adding and subtracting (9) to (8) we obtain the servo-robot dynamics in extended velocity coordinates $S_r = \dot{q} - \dot{q}_r$ as

$$K_{PS}^{-1}\left(H_{SR}(q)\dot{S}_r + C_{SR}(q,\dot{q})S_r + \Upsilon BS_r\right) = u - \tau_\Gamma - Y_{r2}\Theta_2. \tag{10}$$

Model (10) will be useful for stability analysis.

Remark 1: Servo-robot model (6) inherits the well known properties of robot manipulators dynamics (1) listed in [4, Ch. 4].

2.5. Neural approximation property

Consider a smooth function $f(x)$, then for a neural network with sufficient neurons in the hidden layer, there exist an activation function and weights such that

$$f(x) = W^T\varphi(x) + \epsilon, \tag{11}$$

that is, the function $f(x)$ can be approximated by a neural network, where W are ideal, constant and bounded weights with $\|W\|_F \leq W_M$, φ a basis [31], and ϵ the bounded neural network approximation error [32], i.e., $\|\epsilon\| \leq \epsilon_M$. Because the ideal weights W are unknown, the approximation (output) of the implemented neural network is defined as

$$\hat{f} = \hat{W}^T\varphi(x), \tag{12}$$

where $\hat{W}$ is the estimation of the ideal weights W. It is not straight forward to define a basis, however by defining the NN as

$$\hat{f} = \hat{W}\sigma(V^T\chi), \tag{13}$$

with $\sigma(\cdot)$ the sigmoid function, V fixed weights randomly selected, χ the input vector, and $\sigma(V^T\chi)$ is a basis [33]. Fig. 3 shows the representation of the neural network of (13).

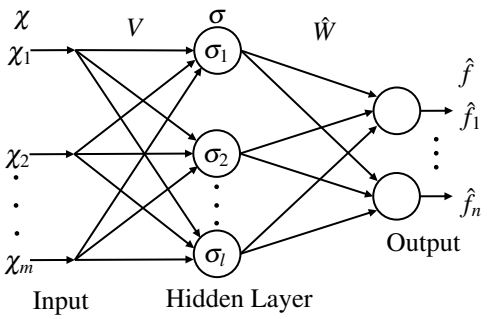


Fig. 3. Neural network architecture.

2.6. Problem statement

The problem is to design the control input u and the adaptation of the weights $\hat{W}$, without knowledge of the system's dynamics, to approximate and compensate for the nonlinear dynamics and guarantee the convergence of the tracking errors.

3. PROPOSED NEUROCONTROLLER

3.1. Neural network adaptation

The objective of the neural network $\hat{f} = \hat{W}^T \sigma(V^T \chi)$ is to approximate the nonlinear dynamics $Y_{r2}\Theta_2 = W^T \sigma(\cdot) + \epsilon$, where $\hat{W} \in \mathbb{R}^{l\times n}$ are adaptive weights, $\sigma \in \mathbb{R}^{l\times 1}$ is the vectorial form of sigmoid activation function, $V \in \mathbb{R}^{5n+1\times l}$ are fixed weights, and $\chi = (1,\ \Delta q,\ \Delta\dot{q},\ q_d,\ \dot{q}_d,\ \ddot{q}_d)^T \in \mathbb{R}^{5n+1\times 1}$ is the input vector to the neural network, with $q_d,\ \dot{q}_d,\ \ddot{q}_d \in \mathbb{R}^n$ the desired position, velocity and acceleration, respectively, and $\Delta q = q - q_d$, $\Delta\dot{q} = \dot{q} - \dot{q}_d$ the position and velocity errors. In this regard, we need to define the adaptation $\dot{\hat{W}}$ to guarantee $\hat{f} \to Y_{r2}\Theta_2$, more precisely, $\hat{W} \to W + \epsilon$, being ϵ the bounded error approximation. Then, the adaptation of the neural network's weights is defined as

$$\dot{\hat{W}} = -\Lambda\sigma(\cdot)S_r^T - \xi\Lambda\hat{W}, \tag{14}$$

where $\Lambda \in \mathbb{R}^{l\times l}$ is a positive definite gain matrix, $\xi \in \mathbb{R}$ is a positive gain, and $S_r \in \mathbb{R}^n$ are the extended velocity error coordinates that will be defined in the next section.

3.2. Controller

The proposed Neuro-Integral Sliding Mode control for the servo-robot system (6) is defined as follows:

$$u = -K_d S_r + \hat{f}, \tag{15}$$

where $K_d = K_d^T \in \mathbb{R}^{n\times n} > 0$ is a gain matrix. Then, the extended velocity error is defined as

$$S_r = S_q + K_i\zeta, \tag{16}$$

where $K_i \in \mathbb{R}^{n\times n}$ is a positive definite diagonal matrix, $S_q = S - S_d$, $S = \Delta\dot{q} + \alpha\Delta q$, and $S_d = S(t_0)e^{-\kappa t}$.

Remark 2: Note that $S_r = \dot{q} - \dot{q}_r$, therefore, to obtain (16) the nominal reference is defined as $\dot{q}_r = \dot{q}_d - \alpha\Delta q + S_d - K_i\int_0^t \mathrm{sgn}(S_q)dt$.

Consider the following Lyapunov candidate function

$$V = \frac{1}{2}S_r K_{PS}^{-1} H_{SR}(q) S_r + \frac{1}{2}tr(\tilde{W}^T\Lambda^{-1}\tilde{W}), \tag{17}$$

with $\tilde{W} = W - \hat{W}$. Taking the derivative along the trajectories we obtain

$$\dot{V} = S_r^T K_{PS}^{-1} H_{SR}(q)\dot{S}_r + S_r^T K_{PS}^{-1}\dot{H}_{SR}S_r - tr(\tilde{W}^T\Lambda^{-1}\dot{\hat{W}}). \tag{18}$$

Using dynamics (10) in (18) leads to

$$\begin{aligned}\dot{V} &= S_r^T K_{PS}^{-1}(-C_{SR} + \frac{\dot{H}_{SR}}{2})S_r + S_r^T\left(u - \tau_\Gamma - Y_{r2}\Theta_2\right)\\ &\quad - S_r^T K_{PS}^{-1}\Upsilon B S_r - tr(\tilde{W}^T\Lambda^{-1}\dot{\hat{W}})\\ &= -S_r^T K_{PS}^{-1}\Upsilon B S_r + S_r^T\left(u - \tau_\Gamma - Y_{r2}\Theta_2\right)\\ &\quad - tr(\tilde{W}^T\Lambda^{-1}\dot{\hat{W}}),\end{aligned} \tag{19}$$

where the skew-symmetric property of the matrix $-C_{SR} + \dot{H}_{SR}/2$ was used. Then, substituting the controller (15) and the weights' adaptation law (14), recalling that $\hat{f} = \hat{W}^T\sigma$ and $Y_{r2}\Theta_2 = W^T\sigma + \epsilon$, and reducing terms we obtain

$$\begin{aligned}\dot{V} &= -S_r^T K_{PS}^{-1}\Upsilon B S_r + S_r^T\left(-K_d S_r - \tilde{W}^T\sigma - \epsilon - \tau_\Gamma\right)\\ &\quad - tr(\tilde{W}^T(-\sigma S_r^T - \xi\hat{W})\\ &= -S_r^T K_{PS}^{-1}(\Upsilon B + K_d)S_r + S_r^T\left(-\epsilon - \tau_\Gamma\right)\\ &\quad + \xi tr(\tilde{W}^T\hat{W}).\end{aligned} \tag{20}$$

Using $\hat{W} = W - \tilde{W}$ leads to

$$\begin{aligned}\dot{V} &= -S_r^T K_{PS}^{-1}(\Upsilon B + K_d)S_r - \xi tr(\tilde{W}^T(\tilde{W} - W)) +\\ &\quad + S_r^T\left(-\epsilon - \tau_\Gamma\right).\end{aligned} \tag{21}$$

Using the bound $tr(\tilde{W}^T(W - \tilde{W})) \le \|\tilde{W}\|_F\|W\|_F - \|\tilde{W}\|_F$, and $\|\tau_\Gamma\| \le \epsilon_\Gamma$, completing squares, and defining $\Pi = K_{PS}^{-1}(\Upsilon B + K_d)$, we obtain

$$\begin{aligned}\dot{V} &\le -\|S_r\|^2\lambda_m(\Pi) - \xi(\|\tilde{W}\|_F - W_M/2)^2 +\\ &\quad + \|S_r\|(\epsilon_M + \epsilon_\Gamma) + \xi\frac{W_M}{4}^2,\end{aligned} \tag{22}$$

with $\lambda_m(\cdot)$ being the minimum eigenvalue, and $\|W\|_F \le W_M$, with W_M a bound. Then for large enough feedback K_d such that $\lambda_m(\Pi) > (\epsilon_M + \epsilon_\Gamma) + \xi W_M^2/4$ and the condition $\|\tilde{W}\|_F > W_M/2$ holds, we obtain that $\dot{V} < 0$ outside a compact set, implying that $S_r \to \epsilon_{Sr}$ and $\tilde{W} \to \epsilon_{\tilde{w}}$ as $t \to \infty$. The boundedness of S_r implies the boundedness of $\dot{S}_r$, that is, $\|\dot{S}_r\| \le \Omega$ [34].

To prove the existence of a sliding mode consider the function $V_{SM} = \frac{1}{2}S_q^T S_q$, whose time derivative is $\dot{V}_{SM} = S_q^T\dot{S}_q$. From (16) we have that $\dot{S}_q = \dot{S}_r - K_i\int_0^t \mathrm{sgn}(S_q)$, using this equation in $\dot{V}_{SM}$, we obtain

$$\dot{V}_{SM} = S_q^T(-K_i\mathrm{sgn}(S_q) - \dot{S}_r) \tag{23}$$

$$\le -\lambda_m(K_i)\,|S_q| + |S_q|\,|\dot{S}_r| \tag{24}$$

$$\le -(\lambda_m(K_i) - \Omega)|S_q|. \tag{25}$$

One can always select K_i such that $\lambda_{min}(K_i) > \Omega$, guaranteeing the existence of a sliding mode at $S_q = 0$. Thus, exponential convergence of the tracking errors is guaranteed, i.e., $(\Delta\dot{q},\ \Delta q) \to (0,\ 0)$ as $t \to \infty$.

4. SIMULATIONS

Simulations are carried out in MATLAB-Simulink 2024. A 3-DoF robot is considered. The constant servo

Table 1. Actuators parameters (for $i = 1, 2, 3$).

Parameter	Value	Unit
k_i	0.0107	Nm/A
k_{Pi}	32	Nm/rad
J_i	79.6×10^{-6}	Nms^2/rad
R_i	6.5	Ω
b_{0i}	4.1×10^{-3}	Nms/rad
k_{bi}	0.006705	V/(rad/s)

parameters are shown in Table 1, and the robot parameters are links mass $m_1 = 7.2$ kg, $m_2 = 5$ kg, $m_3 = 1.9$ kg, links lengths $l_1 = 0.5$ m, $l_2 = 0.5$ m, $l_3 = 0.35$ m, and links inertia $I_1 = 0.015$, $I_2 = 0.02$, $I_3 = 0.016$. The initial conditions are $q(0) = [\pi/4, 0, 0]^T$, $\dot{q}(0) = [0, 0, 0]^T$. The desired trajectory is defined as $q_d = (\cos(tw), \sin(tw), \cos(tw))$, with $w = 0.5$. Without loss of generality, the servos internal gear ratio is defined as $\eta_{mi} = 1$, and the disturbance torque is given by $\tau_\Gamma = \varpi((1+0.1\sin(at)), (1+0.3\sin(at)), (1+0.2\sin(at)))^T$ with $\varpi = 0.5 + 0.5\tanh(10(t-6))$and $a = 3$, corresponding to a smooth but fast change of the disturbance in $t \approx 6$ s.

4.1. Neural network structure, adaptation parameters and feedback gains

The neural network is composed by one input layer, one hidden layer and one output layer, see Fig. 3. The input is defined as $\chi = [1, \Delta q, \Delta\dot{q}, q_d, \dot{q}_d, \ddot{q}_d]^T$, for the 3-DoF robot $\chi \in \mathbb{R}^{16\times 1}$, and the hidden layer is conformed with 10 neurons with the bipolar sigmoid activation function. The constant weights V were selected randomly between 0 and 1, and similarly the initial adaptive weights $\hat{W}(0)$. The gains used in the weight's adaptation law are $\Lambda = I_{10\times 10}1 \times 10^5$, and $\xi = 0.045$. The controller gains were selected as $K_d = diag(200\ 250\ 60)$, $\alpha = diag(3\ 3\ 3)$, $K_i = diag(0.3\ 0.8\ 0.3)$, $\kappa = 10$.

4.2. Results

To notice the contribution of the neural network in the control law, in addition to the proposed neuro-controller, simulations considering only the ISM control were performed. The results of the proposed Neuro-ISM are shown in Fig. 4, while the results of the ISM with the same control gains as the Neuro-ISM are shown in Fig. 5. Figs. 4(a) and 5(a) depict the position error behavior for the Neuro-ISM and ISM, respectively.

Notice that the transient response of the integral sliding mode is more pronounced compared to that of the Neuro-ISM. Furthermore, after the disturbances take effect, at $t \geq 6$, the integral sliding mode continues to exhibit small errors, whereas the Neuro-ISM effectively compensates for the disturbances and drives the errors to zero. The behavior of the velocity errors are shown in Figs. 4(b) and 5(b), notice that after the disturbance, the Neuro-ISM effectively deals with the disturbance, unlike the ISM, that

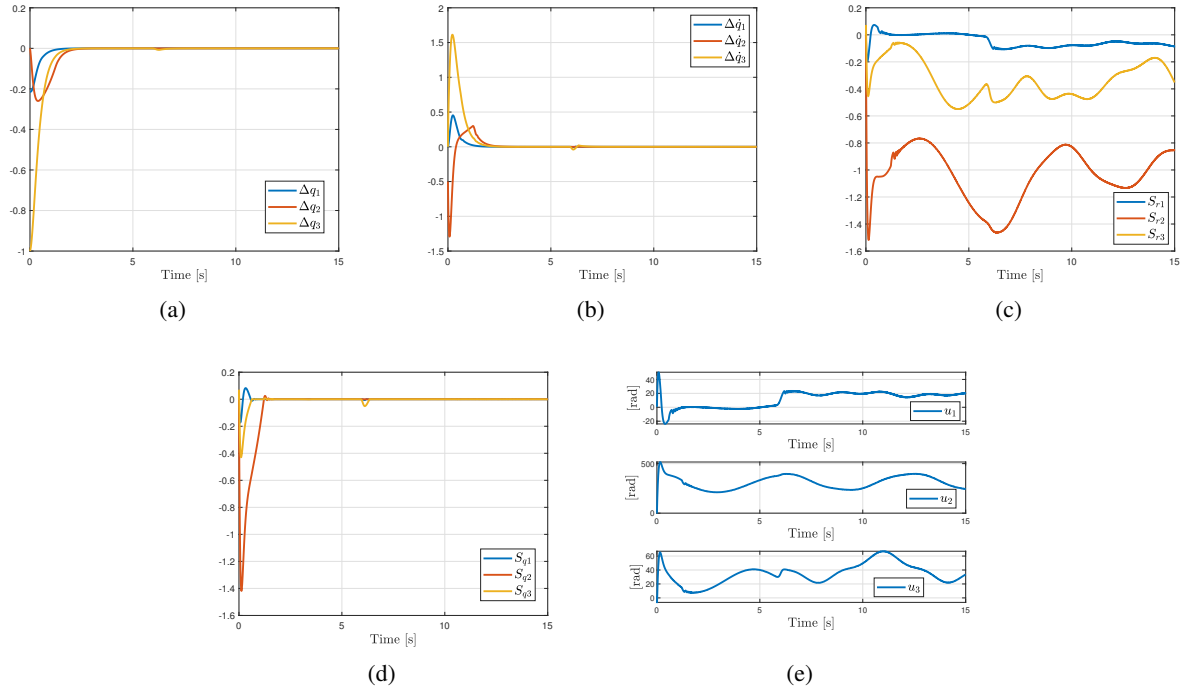


Fig. 4. [NEURO-ISM] Results from the Neuro-ISM control. (a) Position and (b) velocity tracking errors. (c) Extended velocity error and (d) sliding surface. (e) Control signals.

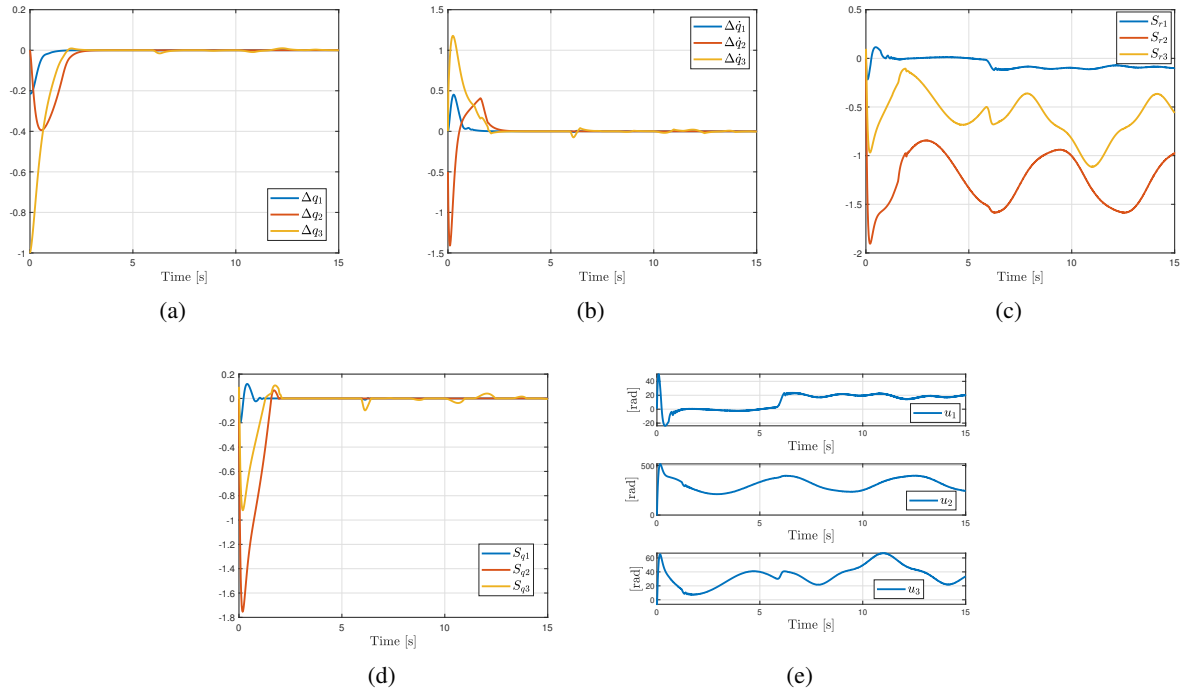


Fig. 5. [ISM] Results from the integral sliding mode. (a) Position and (b) velocity tracking errors. (c) Extended velocity error and (d) sliding surface. (e) Control signals.

presents slight variations. In both controllers, the extended velocity error remains bounded; see Figs. 4(c) and 5(c). However, comparing the sliding surfaces, the integral sliding mode struggles to stay in the surface, see Fig. 5(d), while the Neuro-ISM achieves to maintain the sliding surface at the origin, even after the presence of disturbances, see Fig. 4(d). With respect to the resultant control signal, both have very similar behavior; see Figs. 4(e) and 5(e).

It is worth recalling that the comparison was made with the same conditions and feedback gains. Selecting sufficiently high gains for the integral sliding mode controller must be enough to obtain a better response of the tracking errors. However, this is precisely the contribution of the neural network, by compensating the nonlinear dynamics it is not necessary to select "too large" gains, as in the non-neuro integral sliding mode controller.

The performance indices used to compare the controllers are $ITAE_{pos} = \int_{t_0}^{t} t \, \|\Delta q\|^2$, $ITAE_{vel} = \int_{t_0}^{t} t \, \|\Delta \dot{q}\|^2$, and $IAC = \int_{t_0}^{t} \|\tau\|$, which are shown in Table 2, where our proposal obtained smaller values representing a better performance under those criteria.

Table 2. Neuro-ISM vs ISM mode with identical gains. The best-performing metrics corresponding to our proposal are shown in bold.

Control law	Performance criterion		
	$ITAE_{pos}$	$ITAE_{vel}$	IAC
Our proposal	**0.0924**	**0.4582**	**4703**
Integral sliding mode	0.2241	0.5868	4734

5. CONCLUSIONS

In this work, a neuro-controller is proposed to control a novel dynamic model of robot manipulators equipped with position servo actuators. A low dimensional neural network is used to approximate the nonlinear dynamics of the servo-robot. Stability of the whole system is proven via Lyapunov stability analysis where the boundedness of the closed-loop system trajectories is guaranteed, including the evolution of the neural networks weights. Convergence of the tracking errors is achieved without too large gains, in contrast to ISM control, due to the approximation and compensation of the neural network. Comparative simulation results demonstrate the effectiveness of the neuro-controller despite the presence of bounded and continuous disturbances.

DECLARATIONS

Conflict of Interest

The authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

Authors' Contributions

Luis Pantoja-Garcia: Conceptualization, Writing – original draft, Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Gabriela Zepeda: Conceptualization, Writing – original draft, Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation.

Oscar A. López-Nuñez: Writing – review & editing, Visualization, Validation, Supervision, Resources, Investigation, Data curation.

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REFERENCES

[1] M. Coreless and G. Leitmann, "Continuous state feedback guaranteeing uniform ultimate boundedness for uncertain dynamic systems," *IEEE Transactions on Automatic Control*, vol. 25, no. 5, pp. 1139-1144, 1981.

[2] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*, John Wiley & Sons, New Jersey, 2020.

[3] J.-J. E. Slotine and W. Li, "On the adaptive control of robot manipulators," *The International Journal of Robotics Research*, vol. 6, no. 3, pp. 49-59, 1987.

[4] R. Kelly, V. Santibáñez, and A. Loría, *Control of Robot Manipulators in Joint Space*, Springer, London, 2005.

[5] K. S. Narendra and K. Parthasarathy, "Identification and control of dynamical systems using neural networks," *IEEE Transactions on Neural Networks*, vol. 1, no. 1, pp. 4-27, 1990.

[6] J.-M. Wu, Z.-H. Lin, and P.-H. Hsu, "Function approximation using generalized adalines," *IEEE Transactions on Neural Networks*, vol. 17, no. 3, pp. 541-558, 2006.

[7] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Mathematics of Control, Signals and Systems*, vol. 2, no. 4, pp. 303-314, 1989.

[8] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Networks*, vol. 2, no. 5, pp. 359-366, 1989.

[9] F. Lewis, S. Jagannathan, and A. Yesildirak, "Neural Network Control of Robot Manipulators and Non-linear Systems," *CRC press*, USA, 2020.

[10] K. El Hamidi, M. Mjahed, A. El Kari, and H. Ayad, "Adaptive control using neural networks and approximate models for nonlinear dynamic systems," *Modelling and Simulation in Engineering* , vol. 2020, no. 1, 8642915, 2020.

[11] L. Jin, S. Lin, J. Yu, and J. He, "Robot manipulator control using neural networks: A survey," *Neurocomputing*, vol. 285, pp. 23-24, 2018.

[12] F. L. Lewis and K. G. Vamvoudakis, *Neuro-inspired Control*, in J. Baillieul and T. Samad, eds., Encyclopedia of Systems and Control, Springer, London, 2020.

[13] F. L. Lewis and S. S. Ge, *Neural Networks in Feedback Control Systems*, John Wiley & Sons, Chap. 9, USA, 2005.

[14] F. L. Lewis, "Neural networks for feedback control of robots and dynamical systems," *Proc. of IEEE International Joint Conference on Neural Networks*, vol. 5, 2005.

[15] S. Zhang, Y. Dong, Y. Ouyang, Z. Yin, and K. Peng, "Adaptive neural control for robotic manipulators with output constraints and uncertainties," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 29, no. 11, pp. 5554-5564, 2018.

[16] Q. Liu, D. Li, S. S. Ge, R. Ji, Z. Ouyang, and K. P. Tee, "Adaptive bias rbf neural network control for a robotic manipulator," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 447, pp. 213-223, 2021.

[17] M. Elsisi, K. Mahmound, M. Lehtonen, and M. M. F. Darwish, "An improved neural network algorithm to efficiently track various trajectories of robot manipulator arms," *IEEE Acess*, vol. 9, pp. 11911–11920, 2021.

[18] A. Liu, H. Zhao, T. Song , Z. Liu, H. Wang, and D. Sun, "Adaptive control of manipulator based on neural network," *Neural Computing & Applications*, vol. 33, no. 9, pp. 4077–4085, 2021.

[19] C. Yang, D. Huang, W. He, and L. Cheng, "Neural control of robot manipulators with trajectory tracking constraints and input saturation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 9, pp. 4231–4242, 2021.

[20] G. D. Khan, "Adaptive neural network control framework for industrial robot manipulators," *IEEE Access*, vol. 12, pp. 63477–63483, 2024.

[21] F. Fan, C. Yang, B. Li, and Y. Li, "Neuro-adaptive-based fixed-time composite learning control for manipulators with given transient performance," *IEEE Transactions on Cybernetics*, vol. 54, no. 12, pp. 7668–7680, 2024.

[22] V. T. Yen, W. Y. Nan, P. V. Cuong, N. X. Quynh, and V. H. Thich, "Robust adaptive sliding mode control for industrial robot manipulator using fuzzy wavelet neural networks," *International Journal of Control, Automation, and Systems*, vol. 15, no. 6, pp. 2930–2941, 2017.

[23] R.-D. Xi, X. Xiao, T.-N. Ma, and Z.-X. Yang, "Adaptive sliding mode disturbance observer based robust control for robot manipulators towards assembly assistance," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, 2022.

[24] T. N. Truong, A. T. Vo, and H.-J. Kang, "A model-free terminal sliding mode control for robots: Achieving fixed-time prescribed performance and convergence," *ISA Transactions*, vol. 144, pp. 330–341, 2024.

[25] T. Li, G. Zhang, T. Zhang, and J. Pan, "Adaptive neural network tracking control of robotic manipulators based on disturbance observer," *Processes*, vol. 12, no. 3, 2024.

[26] J. G. Elkins and F. Fahimi, "Online neural sliding mode control with guaranteed stability," *International Journal of Control*, vol. 98, no. 2, pp. 249–259, 2025.

[27] R. García-Rodriguez, and V. Parra-Vega, "Tracking control of robot manipulators using second order neuro sliding mode," *Latin American Applied Research*, vol. 39, no. 4, pp. 285–294, 2009.

[28] G. Zepeda, R. Kelly, and C. Monroy, "A theoretical conversion of torque controllers for handling a position servo-actuated robot manipulator model," *International Journal of Control, Automation, and Systems*, vol. 22, no. 4, pp. 1301-1312, 2024.

[29] R. Firoozian, *Servo Motors and Industrial Control Theory*, Springer, 2014.

[30] M. Scarpino, *Motors for Makers: A Guide to Steppers, Servos, and Other Electrical Machines*, Que Publishing, 2015.

[31] N. Sadegh, "A perceptron network for functional identification and control of nonlinear systems," *IEEE Transactions on Neural Networks*, vol. 4, no. 6, pp. 982–988, 1993.

[32] A. Barron, "Universal approximation bounds for superpositions of a sigmoidal function," *Transactions on Information Theory*, vol. 39, no. 3, pp. 930–945, 1993.

[33] B. Igelnik and Y.-H. Pao, "Stochastic choice of basis functions in adaptive function approximation and the functional-link net," *IEEE Transactions on Neural Networks*, vol. 6, no. 6, pp. 1320–1329, 1995.

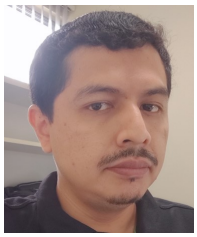
[34] V. Parra-Vega, S. Arimoto, Y.-H. Liu, G. Hirzinger, and P. Akella, "Dynamic sliding PID control for tracking of robot manipulators: Theory and experiments," *IEEE Transactions on Robotics and Automation*, vol. 19, no. 6, pp. 967–976, 2003.



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